

Extracorporeal Therapies for Drug Overdose and Poisoning

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Whether intentional or accidental, poisoning and drug overdose are common medical emergencies that account for around 160,000 presentations annually in the United Kingdom and many more consultations with primary care and National Health Service (NHS) advice services.¹ Most exposures do not require hospitalization, with around 66% of US cases in 2019 being managed in a non-health care facility, usually at the site of the incident.² Overall exposure-related mortality remains relatively low, with 1411 exposure-related fatalities recorded in around 2.15 million exposures in this same US report.² However, poisoning is a significant cause of death in young people.³

The spectrum of agents ingested is wide, ranging from overdose of prescribed and nonprescribed medicinal drugs to poisoning with nonpharmacologic substances and recreational drugs. The pattern of toxin ingestion has changed over the years and varies according to geographic location. Frequently implicated agents in industrialized societies include analgesics (acetaminophen, opioids, and salicylates), antidepressants, sedatives, and antipsychotics. Pesticides are a frequent cause of poisoning in low- and middle-resource countries.³ Legislation has been used to reduce the availability of potential toxins; for instance, paraquat has not been sold in the United Kingdom since July 2008. Similarly, UK legislation limiting pack size of paracetamol (acetaminophen) in 1998 was followed by a significant reduction in the number of deaths from paracetamol overdose.⁴

Effective management of poisoning depends on hemodynamic, respiratory, and other supportive care, prevention of further poison absorption (through oral activated charcoal in specific patients), neutralization of toxicity (e.g., with intravenous *N*-acetylcysteine after significant acetaminophen overdose or digoxin immune Fab [Digibind]), and enhancement of endogenous toxin elimination (e.g., through urinary alkalinization). These aspects of management are well described by resources such as Toxbase in the United Kingdom (<https://www.toxbase.org>) and via the American Association of Poison Control Centers (<https://www.aapcc.org>). Other local and regional poison information services are linked through the website of the European Association of Poisons Centres and Clinical Toxicologists (<https://www.eapcc.org>).

Extracorporeal treatment (ERCT) for poison elimination includes kidney replacement therapy (KRT) and other modalities such as hemoperfusion and therapeutic plasma exchange. However, ERCT is only occasionally needed to manage poisoning and overdose; in the United States in 2019, only around 3400 patients were treated, mostly with hemodialysis (HD) or continuous KRT (CKRT).² Only 21 patients were recorded as receiving hemoperfusion, reflecting a long-term decline explained partly by the increasing rarity of theophylline and barbiturate overdose (historically, the main indications for this technique) but also by the increased use of high-efficiency, high-flux HD, allowing removal of toxins previously regarded as being nondialyzable.^{5,6}

ERCT has long been used to manage poisoning,⁵ but common practices are often not supported by high-quality evidence, at least partly because of the practical and ethical barriers to randomized controlled trials.⁵ Despite these barriers the EXTRIP (Extracorporeal Treatments in Poisoning) consensus group (<https://www.extrip-workgroup.org>)⁶ has created several useful resources for clinicians endorsed by various national and international stakeholder societies.

WHEN SHOULD EXTRACORPOREAL REMOVAL BE CONSIDERED?

ERCT is most commonly used for lithium, toxic alcohol, and salicylate ingestions⁷; other potential indications are shown in Table 103.1. Extracorporeal elimination should be considered when it is evident that other treatments (e.g., activated charcoal, specific antidotes) will not reduce poison levels below toxic thresholds. It is therefore important to understand the general principles underpinning the use of ERCT to make decisions about management of poisonings, particularly for those for which no consensus or evidence base exists.

Extracorporeal removal is appropriate for only a small proportion of poisonings. The potential advantages of therapy must be balanced with the risks that are generic to all forms of ERCT (e.g., acute transfer to a specialist center, vascular access, anticoagulation^{8,9}) and those specific to its use in poisoning (discussed later under individual treatment modalities). Factors that help determine the potential value of extracorporeal removal are summarized in Box 103.1. It should be noted that effective clearance by ERCT does not necessarily prevent ongoing end organ toxicity⁵ or death, as may occur in cases of paraquat poisoning.¹⁰

Molecular Weight

Most ingested poisons have a molecular weight (MW) in the 100 to 1000 Da range and so are amenable to removal by HD,¹¹ which is the modality of choice for clearance of small solutes. Higher flux membranes extend the range of dialyzable solutes up to approximately 10,000 Da, particularly if performed for long hours, and allow removal of toxins such as theophylline and carbamazepine, for which hemoperfusion might historically have been indicated.¹¹ Molecules toward the upper end of this range and up to about 40 kDa are more effectively removed using modalities with a significant convective component such as hemofiltration (HF) or hemodiafiltration (HDF). Even larger solutes may be removed by non-KRT, such as plasma exchange or one of the albumin dialysis techniques used for liver support.

Protein Binding

Diffusive and convective modalities (HD, HF, HDF; either intermittent or continuous) are generally unsuitable for removing poisons with protein binding greater than 80%.¹¹ However, certain toxins (e.g.,

TABLE 103.1 Poisonings for Which Extracorporeal Removal May Be Indicated*

Agent	Preferred Modality	Other, Acceptable Modalities
Acetaminophen	IHD	IHP, CKRT, Ex
Long-acting barbiturates	IHD	HP, CKRT
Carbamazepine	IHD	IHP, CKRT
Ethylene glycol	IHD	CKRT
Lithium	IHD	CKRT ^b
Metformin	IHD	CKRT ^b
Methanol	IHD	CKRT
Phenytoin	IHD	IHP
Salicylates	IHD	IHP, CKRT, Ex
Thallium	IHD	IHP, CKRT
Theophylline	IHD	HP, CKRT, Ex
Valproate	IHD	IHP, CKRT

Other agents that may be amenable to removal by extracorporeal techniques include β -blockers, meprobamate, deferoxamine, aminoglycosides (with high flux membranes), pentoxifylline, sodium edetate, *Amanita phalloides* toxin, and paraquat.

*After initial treatment, both IHD and CKRT are equally acceptable.

CKRT, Continuous kidney replacement therapy; Ex, exchange transfusion in neonates; IHD, intermittent hemodialysis; IHP, intermittent hemoperfusion.

See EXTRIP. Blood Purification in Toxicology: Reviewing the Evidence and Providing Recommendations. <http://www.extrip-workgroup.org/>.

BOX 103.1 Factors Affecting Toxin Removal by Extracorporeal Therapy

- Molecular weight
- Protein binding
- Volume of distribution
- Solute compartmentalization
- Contribution of extracorporeal toxin removal relative to endogenous clearance

salicylate, valproate) are highly protein bound under normal circumstances but saturate protein-binding sites during poisoning, leading to high (and therefore dialyzable) serum concentrations of free agent.⁵ In addition, easy dissociation of protein-toxin complexes may allow significant diffusive removal, as with phenytoin.⁵ Agents with very high (>90%) protein binding and that are not amenable to diffusive or convective elimination may be removed by plasma exchange (e.g., L-thyroxine and cisplatin).¹¹

The interplay between MW and protein binding and their impact on modality choice are shown in Fig. 103.1.

Volume of Distribution

The efficacy of toxin removal is also influenced by the theoretical volume of distribution (V_D). Substances confined to the bloodstream will have a low V_D (~0.07 L/kg body weight); those distributed in the extracellular space, a V_D of approximately 0.2 L/kg; and those confined to total body water, approximately 0.6 L/kg. Higher distribution volumes are often found in substances with avid tissue binding, lipophilicity, or sequestration. As V_D increases, more solute must be removed for achievement of a particular blood level. The ideal solute for extracorporeal removal would therefore have a low distribution volume and would be located within a single, well-mixed compartment that

is directly accessible by ERCT. There are exceptions: carbamazepine, metformin, and thallium all have high distribution volumes but are amenable to extracorporeal removal.¹¹

Solute Compartmentalization

Solutes are often distributed across one or more body compartments that may not be directly accessible by ERCT (Fig. 103.2). If there is any resistance to solute movement between the accessible proximal and the remote compartments, disequilibrium will develop during ERCT, reducing the overall efficiency of toxin removal, and potentially leading to a large postsession rebound in toxin blood levels. Potentially toxic posttreatment levels may be missed if the immediate postdialysis blood level is used to estimate elimination (Fig. 103.3). After ERCT for lithium poisoning, for example, it is recommended that levels be rechecked after 6 to 12 hours to detect clinically significant rebound.¹² Vigilance for rapidly rising levels is also important if ongoing toxin absorption is suspected and may require earlier and more frequent monitoring.

High clearance techniques (i.e., intermittent HD or HDF) rapidly reduce blood solute levels but are more likely to result in intercompartmental disequilibrium. Extending sessions beyond 4 hours can ameliorate rebound, but intermittent HD and the diffusive component of intermittent HDF are inherently inefficient processes that depend on the solute concentration presented to the dialyzer. Most solute removal will therefore occur at the start of dialysis, so any gains in solute removal will be disproportionately lower as dialysis time extends. Increasing treatment frequency can mitigate the risk for significant rebound to toxic blood levels.

Compartmentalization need not preclude HD-/HDF-mediated toxin removal, but the total body burden of highly compartmentalized toxins (e.g., digoxin; tricyclic antidepressants [TCAs]) cannot be effectively reduced by HD even though blood levels are rapidly reduced. Therefore, ERCT is not recommended in digoxin or TCA poisonings⁶ despite occasional reports of successful clearance of TCAs with dialysis.⁵

Contribution of Extracorporeal Toxin Relative to Endogenous Clearance

Even toxins that may be readily cleared by ERCT may not require such treatment if endogenous clearance rates are high. It is proposed that ERCT is justified if it increases total body clearance by at least 30%¹³ or if endogenous clearance is less than 4 mL/min/kg body weight.¹⁴ Unfortunately, these thresholds can be difficult to estimate at the bedside, although an indication of endogenous clearance rates is available for some drugs (see <https://www.medicines.org.uk/emc/>), and estimates of dialyzer clearances can be obtained from product inserts. For HF, an estimate of clearance can be provided by the ultrafiltration rate. Maximal clearances for ERCT are approximately 240 mL/min (for intermittent HD/HF/hemoperfusion), 50 mL/min (for CKRTs and plasma exchange), and 10 mL/min (for exchange transfusion).¹¹

However, endogenous elimination in the setting of poisoning may be markedly lower than in health. High endogenous clearances of approximately 2000 mL/min (e.g., for cocaine, labetalol, toluene, and verapamil)¹¹ would appear to obviate the need for ERCT. However, if these elimination routes are limited by end organ dysfunction (e.g., by kidney or liver failure), ERCT may be life-saving.

These considerations are summarized in Fig. 103.4.

TREATMENT MODALITIES

The range of ERCT used for toxin elimination with parameters that might be manipulated to enhance toxin clearances is shown

The Impact of Molecular Weight and Protein Binding on Modality Choice

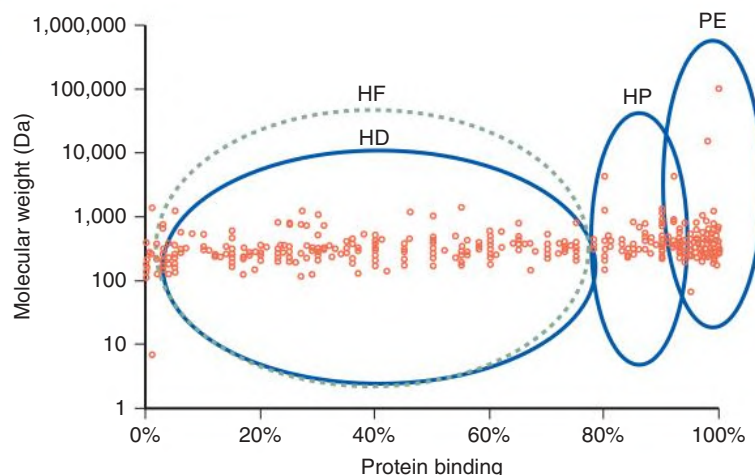


Fig. 103.1 Impact of Molecular Weight and Protein Binding on Modality Choice. Circles indicate for which poisons a specific modality might be most useful. The decline in use of hemoperfusion in many parts of the world means that it may be unavailable as a treatment option. In these circumstances, plasma exchange, which is widely available, may be used. HD, Hemodialysis; HF, hemofiltration; HP, hemoperfusion; PE, plasma exchange. (From Ghannoum M, Roberts DM, Hoffman RS, et al. A stepwise approach for the management of poisoning with extracorporeal treatments. *Semin Dial.* 2014;27[4]:362–370.)

Development of Intercompartmental Solute Disequilibrium

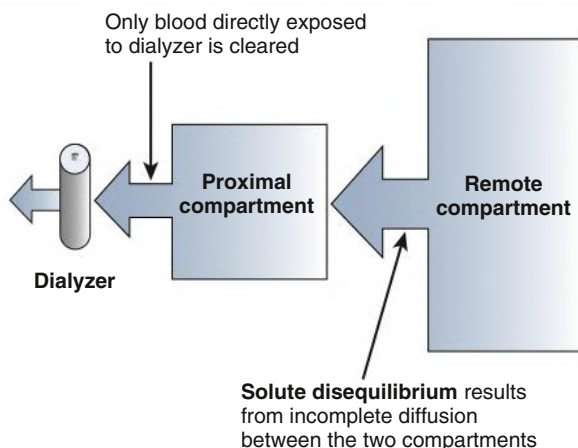


Fig. 103.2 Intercompartmental Solute Disequilibrium. Access of dialyzer to toxin is limited by disequilibrium, which retains toxin in the remote compartment.

in Box 103.2. Generic considerations, including the basics of physical process, nomenclature, and vascular access (and its care) are detailed elsewhere.^{8,9}

Intermittent Hemodialysis, Hemofiltration, and Hemodiafiltration

Diffusion against a steep concentration gradient (as in intermittent HD) encourages the rapid removal of smaller solutes, including most poisons, unless there is significant protein binding. Clearances can be enhanced by increasing dialyzer efficiency (indicated by the KoA, the urea mass transfer area coefficient) or membrane surface area. Larger

Postdialysis Solute Rebound to Toxic Levels

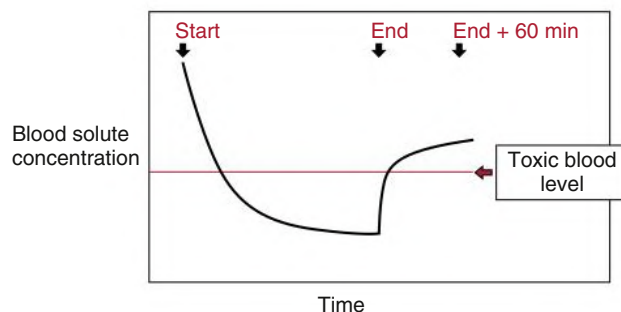


Fig. 103.3 Postdialysis Rebound. Falling solute concentration during dialysis with rapid postdialysis rebound to toxic levels.

solute removal can be enhanced by increasing dialyzer flux (K_{uf}) when intermittent HD is used (for toxins with MW between 500 Da and 10,000 Da, such as deferoxamine and aminoglycosides) or by introducing convective removal (with HF or HDF), which can allow removal of toxins up to approximately 40,000 Da.

Intermittent HD is usually the first-choice extracorporeal modality because of its common availability, the rapidity of toxin removal, and the low MW of common poisons. The role of other kidney replacement modalities is less clear because of a lack of published data.

Peritoneal Dialysis

Peritoneal dialysis (PD) is not an “extracorporeal” technique per se and is rarely used to treat poisoning because of the comparatively slow rate of clearance, the risks associated with acute PD catheter insertion, and the widespread availability of other modalities, at least in the

Stepwise Approach to Initiation of Extracorporeal Therapies in Poisoning

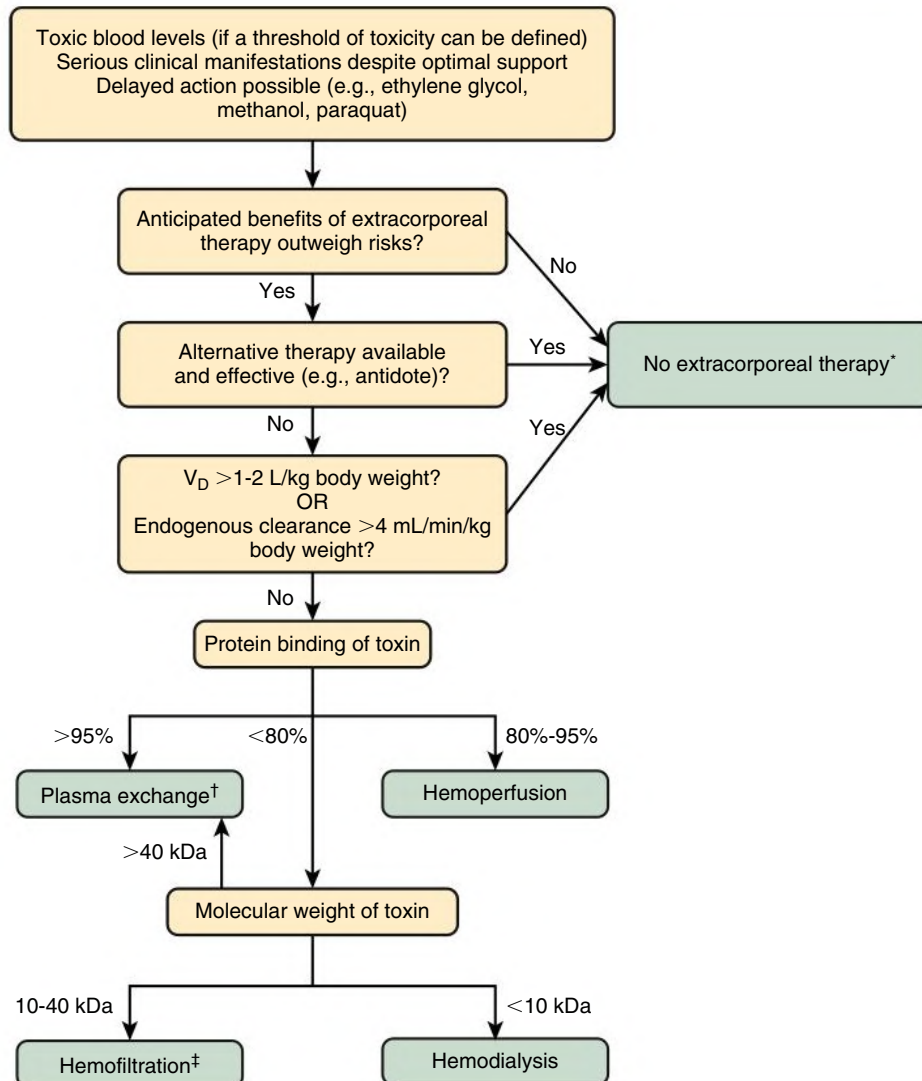


Fig. 103.4 Stepwise Approach to the Initiation of ERCT for Enhanced Elimination in Poisoned Patients.

*Unless other indication (e.g., acute kidney injury [AKI], such as from acetaminophen overdose; severe metabolic acidosis; severe liver failure; hypothermia; hyperthermia). Choice of technique will depend on the indication (e.g., one of the kidney replacement therapies for AKI). †Or albumin dialysis if available. ‡Or hemodialysis with high cut-off membrane, if available (pore size cut-off ~65 kDa). V_D , Solute volume of distribution. (Modified from Ghannoum M, Roberts DM, Hoffman RS, et al. A stepwise approach for the management of poisoning with extracorporeal treatments. *Semin Dial*. 2014;27[4]:362–370.)

industrialized world. It may have a role in the treatment of poisoning in childhood because its lower clearance may be sufficient for the smaller solute distribution volumes in children and because of technical challenges associated with HD in very young children.

Continuous Kidney Replacement Therapy

CKRT may be used when intermittent HD is not immediately available or when more rapid solute removal would be compromised by significant intercompartmental disequilibrium. For small-solute clearances, continuous HF and continuous HD have near kinetic equivalence. Full saturation of dialysate effluent in continuous HD, because of its slow flow rates, gives a small-solute concentration similar to both the ultrafiltrate from HF and plasma water as it leaves the hollow-fiber device.

CKRT gives better solute clearances when applied over the course of several days, but elimination is not as rapid as with intermittent HD. Delivered small-solute clearances of CKRT can be maximized by combining diffusion and convection in continuous HDF. If it is logistically possible, an ideal combination may be initial use of intermittent HD for rapid reduction of toxin levels, with continuous therapy then used to ameliorate any postdialysis rebound when this is anticipated.

Although small-solute clearances are similar in continuous HF and continuous HD, the former should be used in preference to remove larger toxins.

Because HF and HD modalities are often applied for drug elimination in the absence of kidney dysfunction or serious electrolyte disturbance, careful monitoring for evolving biochemical abnormalities is required.

BOX 103.2 Practical Considerations in Prescribing Extracorporeal Therapy for Poisonings

Intermittent Hemodialysis

- Maximize Q_b and dialyzer membrane surface area
- Aim for Q_b/Q_d ratio of 2.5:1 or greater (although diminishing benefits gained from increasing Q_d beyond ~600 mL/min)³³
- Maximize treatment times to at least 4 hours
- Use high-flux (Kuf), high-efficiency (KoA) dialyzer, particularly if middle molecular weight solutes (>1000 Da) are to be cleared

Intermittent Hemofiltration/Filtrative Component of Intermittent Hemodiafiltration

- Aim for postdilution fluid replacement to maximize efficiency with high-flux membrane
- Maximize Q_{uf} according to filtration fraction

Continuous Kidney Replacement Therapies^a

- Aim for high convective clearances (i.e., postdilutional CVVH with high-flux membrane) for larger solutes (>1000 Da)
- Aim for high diffusive clearances (i.e., CVVHD) for small solutes (≤ 1000 Da)
- For CVVH, maximize Q_{uf} according to filtration fraction^b
- For CVVHD, CVVHDF, maximize Q_d up to at least 2.5 L/h^c for small solutes
 - For larger solutes (e.g., of an equivalent size to β_2 -microglobulin; MW 11,800), there may be little gain from a $Q_d > 1.5$ L/h³⁴

Hemoperfusion

- Limit Q_b to ~100 to 250 mL/min (see section on hemoperfusion)
- Change cartridge every 3 to 4 hours
- Consider benefits of charcoal vs. resin cartridges depending on poison¹⁵

Plasma Exchange

- Aim for two plasma volume exchanges per day
- Modify replacement fluid according to poison (see text for details)

^aAiming for conventional effluent flow rates (i.e., 25 mL/kg/h)³⁶ for support of the patient with acute kidney injury may fail to maximize potential poison clearances.

^bThe chief risk with postdilutional fluid replacement is hemoconcentration. This can be minimized by keeping the filtration fraction (FF) at 50% or less. The minimum Q_b to achieve this can be calculated from the formula³⁵: Minimum $Q_b = Q_{uf}/(0.5 \times [1 - \text{Hematocrit}])$. Alternatively, the maximum Q_{uf} would solve as: $Q_{uf} = Q_b \times 0.5 \times (1 - \text{Hematocrit})$. In addition, transmembrane pressures should be kept <400 mm Hg to help prevent membrane fiber rupture.

^cFor small solutes at Q_b 150 mL/min, there appears to be a direct linear relationship between increasing clearances and Q_d to flow rates of at least 2.5 L/h.³⁴ It would therefore seem reasonable to maximize Q_d as machine hardware, dialyzer specifications, and practicalities (fluid availability, nursing workload) allow.

CVVH, Continuous venovenous hemofiltration; CVVHD, continuous venovenous hemodialysis; CVVHDF, continuous venovenous hemodiafiltration; MW, molecular weight; Q_b , extracorporeal blood flow; Q_d , dialysate flow; Q_{uf} , ultrafiltration rate.

From Bouchard J, Roberts DM, Roy L, et al. Principles and operational parameters to optimize poison removal with extracorporeal treatments. *Semin Dial.* 2014;27(4):371–380.

Hypokalemia can be corrected by adjusting the dialysate or replacement fluid composition and by supplementation. Hypophosphatemia can be addressed with standard supplements. At least for intermittent HD, the risk of developing metabolic alkalosis can be attenuated in patients with normal kidney function and baseline serum bicarbonate by using a dialysate bicarbonate concentration at the lower end of the



Fig. 103.5 Charcoal hemoperfusion cartridge.

Extracorporeal Circuit for Hemoperfusion

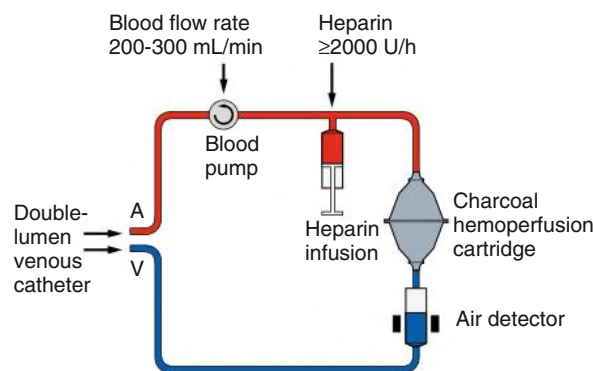


Fig. 103.6 Extracorporeal circuit for hemoperfusion.

deliverable range. Because the bicarbonate concentration of premanufactured replacement or dialysate fluid in CKRT cannot be manipulated, intermittent HD against a low bicarbonate dialysate may be preferable if severe metabolic alkalosis is present.

Hemoperfusion

The technique involves the extracorporeal circulation of blood through a hemoperfusion cartridge (Fig. 103.5) containing an adsorbent material such as activated charcoal or a resin. Hemoperfusion removes substances that bind to the adsorbent material. It is effective at removing uncharged molecules through competitive binding, especially those that are significantly plasma protein bound and lipophilic, although those toxins exceeding about 5000 to 10,000 Da are less well removed.¹⁵ As with other modalities, the decision to start hemoperfusion should be based on its likely proportional contribution to total toxin elimination, as well as the availability of potentially safer techniques such as long-hour, high-flux dialysis (see later).

Other than the cartridge and the lack of dialysate and other replacement fluid circuits, the disposables and hardware for hemoperfusion are the same as those used for intermittent HD (Fig. 103.6). Blood flow should be limited to approximately 100 to 250 mL/min to minimize the risk of hemolysis during passage through the cartridge.¹⁶

Standard anticoagulation protocols may be insufficient for hemoperfusion because heparin is also adsorbed. A larger initial bolus and maintenance dose of unfractionated heparin is usually required (e.g., ≥ 2000 U/h), with adjustments guided by regular monitoring of clotting times during the procedure. The manufacturer's instructions should be reviewed carefully because some cartridges require priming with dextrose solution to prevent hypoglycemia. Other complications

include hypocalcemia, hypophosphatemia, charcoal embolization, leukopenia, thrombocytopenia (usually uncomplicated and self-limiting), pyrogenic reactions, and adsorption of coagulation factors.¹⁵

p0145 The charcoal sorbent particles may be coated with a polymer or other substance to increase biocompatibility and reduce the risk of embolization. Pretreatment of cartridges with albumin, plasma, or heparin to improve biocompatibility and reduce removal of endogenous molecules is of uncertain benefit.¹⁵

p0150 Saturation of the adsorbent material limits the duration of treatment with any individual hemoperfusion cartridge. It is usually recommended that cartridges are changed approximately every 3 to 4 hours, but treatments of this length are usually adequate for a significant lowering of toxin blood levels.¹⁵

p0155 If a toxin is equally removed by intermittent HD and hemoperfusion, intermittent HD should be used preferentially because it is less risky and also allows correction of fluid balance, electrolyte status, and other metabolic abnormalities.

p0160 Decreasing familiarity with hemoperfusion and increasing scarcity of the requisite cartridges, coupled with the ready availability of alternative treatment (long-hour, high-flux kidney support), will likely continue to reduce its application, although the modality remains a more used option in China, India, and elsewhere.¹⁵

s0065 Other Modalities

p0165 Hybrid modalities, such as sustained low-efficiency dialysis, have theoretical appeal for management of poisoning and overdose but require further evaluation.

p0170 There are a few reports of the use of plasma exchange and the molecular adsorbent recirculating system in poisoning. In principle, both should be useful for toxins that are strongly protein bound but not lipid soluble, are tissue bound, or have a large V_D outside the blood compartment.¹⁷ Plasma exchange is most commonly used in *Amanita phalloides* poisoning but may also prevent limb loss in snake envenomation and reduce toxic levels of biologic therapeutic agents (e.g., monoclonal antibodies).¹⁷ Plasma exchange can clear the red cell fragments and free hemoglobin that result from poisoning with sodium chlorate or other agents that can cause hemolysis. The replacement fluid used in plasma exchange warrants some consideration because the plasma free fraction of toxins that are highly albumin bound may rebound after treatment if insufficient albumin is administered.¹⁷ Some toxins have a greater affinity for other plasma proteins such as α_1 -acid glycoprotein than for albumin (e.g., dipyrindamole, quinidine, imipramine, propranolol, and chlorpromazine), arguing for the use of plasma-based rather than albumin-based replacement in these situations.¹⁷ Plasma replacement may be preferable where there is a significant coagulopathy, such as in certain instances of snake envenomation.¹⁷ One to two plasma volume exchanges per day is likely to be sufficient until toxic manifestations and tissue release have subsided.¹⁷

p0175 Exchange transfusion may be useful in neonates and small children who have been poisoned with toxins that have a low volume of distribution (e.g., theophylline, salicylates).¹¹

s0070 EXTRACORPOREAL THERAPY FOR SPECIFIC DRUGS AND POISONS

s0075 Alcohols

p0180 Ethylene glycol (MW 62 Da) and methanol (MW 32 Da) can be found in antifreeze, deicing solutions, and windshield cleaning fluid. The ingestion of as little as 1 g/kg body weight of either methanol or ethylene glycol is potentially lethal. Toxicity results from the metabolism of ethylene glycol and methanol by alcohol dehydrogenase to glycolic acid and formic acid, respectively, and may result in optic nerve

damage, seizures, coma, and ultimately death. Poisoning should be suspected in any patient presenting with nausea, vomiting, abdominal pain, impaired consciousness, severe metabolic acidosis, and acute kidney injury (AKI).

Early recognition of ethylene glycol or methanol poisoning may allow treatment with ethanol or fomepizole to inhibit hepatic alcohol dehydrogenase. Fomepizole has now replaced ethanol as first-line therapy because the former has more predictable pharmacokinetics than ethanol, has a safer side effect profile, shortens ICU and hospital stays, and decreases need for HD, at least in patients poisoned with ethylene glycol.^{18,19}

Clinicians often have to make treatment decisions based on clinical suspicion and without serum drug levels. Fomepizole should be prescribed if there is a clear history of ethylene glycol and/or methanol ingestion, if the osmol gap is greater than 10 mOsm/kg, or the ethylene glycol/methanol blood level is greater than 20 mg/dL (3.2 mmol/L for ethylene glycol and 6.2 mmol/L for methanol). Fomepizole is administered as a loading dose of 15 mg/kg intravenously (IV). This is followed by 10 mg/kg every 12 hours for 4 doses and then 15 mg/kg every 12 hours until the pH is normal and the serum concentration of ethylene glycol and methanol is less than 20 mg/dL (or <10 mg/dL in the presence of end organ damage). Patients who have ingested methanol also should receive folinic acid 50 mg IV every 6 hours to enhance the metabolism of formic acid. Administration of thiamine 100 mg IV or pyridoxine 50 mg IV should be considered in patients with poor nutritional status after ethylene glycol ingestion.

If fomepizole is not available, ethanol can be used. It is a competitive inhibitor of antidiuretic hormone (ADH), as ADH has greater affinity for ethanol than for methanol or ethylene glycol. Ethanol 10% can be administered as a loading dose of 10 mL/kg over 60 minutes to avoid side effects, such as hypotension, respiratory depression, and excessive somnolence. A maintenance infusion should be started at 1 mL/kg/h, aiming to maintain an ethanol concentration of around 100 mg/dL, but an ethanol concentration as low as 20 mg/dL can effectively inhibit the metabolism of methanol or ethylene glycol. Ethanol should be administered until no ethylene glycol or methanol is detectable in the blood. Intravenous sodium bicarbonate should be considered to help correct any associated metabolic acidosis.

HD is considered for treatment of alcohol poisonings to expedite removal of the alcohol and its metabolites. Ethylene glycol and methanol have low MWs, limited protein binding, single-compartment kinetics, and a limited volume of distribution and thus are efficiently cleared by dialysis. Immediate HD is recommended in the presence of severe metabolic acidosis (pH 7.15) regardless of drug level, or if serum concentration is greater than 50 mg/dL (8.1 mmol/L for ethylene glycol and 15.6 mmol/L for methanol) with pH less than 7.3. If no levels are available, HD can also be indicated in the setting of coma seizures or the presence of end organ damage such as AKI or visual disturbance. Because fomepizole is removed by HD, a maintenance infusion of 1 to 1.5 mg/kg/h should be prescribed for the duration of the HD session after administration of the loading dose. Similarly, ethanol doses should be doubled during the dialysis session.

To achieve optimal clearance, the dialyzer should have a large surface area (>1.5 m²) and the blood flow rate should be greater than 300 mL/min. Bicarbonate buffer should be used, and serum concentrations of ethylene glycol and methanol should be measured 2 hours after the treatment to take account of rebound. HD should continue until the pH has normalized and the concentration of ethylene glycol or methanol is less than 25 mg/dL (<4.0 mmol/L for ethylene glycol and <7.8 mmol/L for methanol). Based on their half-life, it is recommended to start with 8 hours for intermittent HD or 18 hours for continuous modalities if there is no knowledge of the starting concentration or

rate of removal of methanol or ethylene glycol. Regardless, acid-base status should be monitored after HD to detect recurrence of poisoning, which may prompt the reinitiation of HD. Patients who have ingested large quantities of ethylene glycol and methanol may require further HD treatment.

p0210 The EXTRIP workgroup recommends intermittent HD as the modality of choice; continuous modalities are acceptable if intermittent HD is not available, although they are not as effective at removing methanol.²⁰

s0080 **β-Blockers**

p0215 β-Blocker overdose will manifest with bradycardia and hypotension and may also include altered mental state, seizures, bronchospasm, hypoglycemia, and cardiogenic shock. Extracorporeal removal is rarely required unless patients have not improved despite maximal medical therapy. β-Blockers have different physicochemical properties and pharmacokinetics that will affect their removal by ERCT. Specifically, ERCT is only effective for hydrophilic, minimally protein-bound drugs such as atenolol, sotalol, nadolol, and acebutolol. Propranolol, carvedilol, labetalol, timolol, and metoprolol are not or minimally removed. Recently, the EXTRIP workgroup recommended against using ERCT for patients severely poisoned with propranolol and offered no recommendation for ERCT in patients severely poisoned with atenolol or sotalol because of apparent balance of risks and benefits, except for impaired kidney function in which ERCT was suggested. Indications for ERCT in patients with impaired kidney function include refractory bradycardia and hypotension for atenolol or sotalol poisoning, and recurrent torsades de pointes for sotalol.²⁰ Intermittent HD is recommended in hemodynamically stable patients. CKRT may be considered in patients who are hemodynamically unstable.

s0085 **Lithium**

p0220 Severe lithium poisoning may manifest with arrhythmias, hypotension, confusion, coma, or seizures. Lithium is a small (MW 7 Da) monovalent cation with properties similar to those of sodium. It is administered as either lithium citrate (liquid formulation) or lithium carbonate (solid formulation) and almost completely absorbed, with peak concentrations occurring in 30 minutes to 2 hours with immediate release capsules or 4 to 5 hours with modified-release preparations.

p0225 Mild lithium toxicity can be present with levels between 1.5 and 2.5 mEq/L and moderate toxicity when greater than 2.5 mEq/L and severe toxicity when greater than 3.5 mEq/L.²¹ However, clinical manifestations are both variable and dependent on the specific pattern of poisoning.²² The delayed symptoms in patients with acute lithium poisoning are due to the slow diffusion of lithium to the brain; thus lithium concentrations are only rough correlates of the potential risk of toxicity and should be interpreted based on history, clinical findings, and kidney function.

p0230 In patient with preserved kidney function, intravenous hydration should be provided with isotonic saline at a rate depending on the patient's fluid status and cardiac function. In patients with lithium-induced nephrogenic diabetes insipidus, sodium levels need to be closely monitored during intravenous hydration to prevent hypernatremia and potentially deteriorating neurologic symptoms. Hemodialysis is recommended if the serum lithium concentration is greater than 4 mmol/L, or greater than 2.5 mmol/L in a patient with central nervous system (CNS) manifestations of toxicity, such as altered mental status and seizures.¹² The protein binding of lithium is negligible, and it is easily removed by HD. High-efficiency HD can achieve a lithium clearance of 180 mL/min—superior to that achieved by native kidney function, which is limited by significant proximal tubular reabsorption. However, impairment of kidney function is often present in

lithium toxicity due to lithium's chronic nephrotoxic effects. Lithium equilibrates relatively slowly between the extracellular and intracellular compartments; thus postdialysis rebound may be significant,²¹ and extended or frequent HD treatment may be needed to minimize its effect. Postdialysis lithium rebound occurs can be pronounced after high-efficiency techniques, reaching 0.5 to 1.0 mEq/L after 6 to 12 hours. Rebound from ongoing absorption can occur in poisonings from extended-release formulations or patients with decreased gastrointestinal motility and may be associated with recurrence of symptoms or clinical deterioration. Serum lithium levels should be checked at 6 hours after dialysis to guide further therapy, and, if the level remains below 1 mEq/L, a second treatment may not be necessary. CKRT may complement intermittent HD by helping mitigate the effect of the postdialysis rebound.

Metformin

Metformin (MW 166 Da) is the most commonly prescribed oral antidiabetic drug. It inhibits gluconeogenesis, facilitates cellular glucose uptake by muscle and adipose cells, and decreases insulin resistance in patients with type 2 diabetes. Metformin toxicity is associated with mortality of 30%, and its management can be challenging. A recent literature review noted that metformin poisoning was the most common toxicologic indication for ERCT.²³ Symptoms include tachypnea, nausea, abdominal pain, tachycardia, hypotension, and, in the setting of AKI, severe lactic acidosis. Metformin has negligible plasma protein binding and a volume of distribution ranging from 63 to 276 L (1–5 L/kg), but the limiting factor for its extracorporeal elimination is the relatively large volume of distribution. Metformin is excreted, unmetabolized, via transporters in the proximal tubules of the kidneys and may accumulate in acute and chronic kidney disease.²⁴ Metformin-associated lactic acidosis (MALA) occurs due to inhibition of hepatic gluconeogenesis from lactate and enhanced conversion of glucose to lactate in the small intestine. The mortality from MALA is high, and treatment options are limited. Medical management involves supportive measures and the intravenous administration of sodium bicarbonate. ERCT is recommended in patients who are unresponsive to medical management.²² As metformin is almost exclusively eliminated by the kidneys, the presence of impaired kidney function will increase the likelihood of severe and more prolonged toxicity. Liver failure can impair lactate clearance, and its presence should also lower the threshold for dialysis support. Extracorporeal removal can be beneficial in metformin toxicity by not only removing the drug but also by delivering a more rapid, predictable, and safe correction of acidemia than can be achieved with bicarbonate therapy, and correction of electrolyte abnormalities and supportive therapy if kidney function is impaired. Both intermittent HD and CKRT can provide effective extracorporeal removal, but a rebound after extracorporeal removal can result in a marked resurgence of lactic acidosis after therapy is discontinued. Close monitoring of the acid-base status is recommended, and extended duration or repeat session may be needed if severe metabolic acidosis (pH ≤ 7.0) is persistent or serum lactate concentration is greater than 20 mmol/L.

Salicylates

Aspirin remains a commonly used analgesic and widely prescribed antiplatelet therapy; despite improvements in supportive care, poisoning by salicylates (MW 180 Da) remains an important cause of poisoning-related mortality.

Aspirin is rapidly absorbed in the stomach, with peak blood concentrations usually reached within 1 hour. At therapeutic levels, 90% of salicylate is protein bound and therefore limited to the vascular space, being metabolized in the liver to salicyluric acid, which is less

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toxic and more rapidly excreted by the kidney. The clinical manifestations of salicylate overdose include fever, sweating, tinnitus, epigastric pain, nausea, vomiting, diarrhea, vertigo, and blurring of vision. Severe overdoses may progress to depression of mental state, noncardiogenic pulmonary edema, and death. Salicylate intoxication initially results in hyperventilation and respiratory alkalosis. This is followed by metabolic acidosis secondary to the accumulation of lactic acid and ketoads. Therefore, patients may present with either respiratory alkalosis or mixed respiratory alkalosis–metabolic acidosis. Diagnosis is based on the presenting history and clinical examination and confirmed with plasma salicylate levels.

p0250 Increased salicylate tissue penetration and toxicity can occur with only small decreases in pH because of increased concentration of non-ionized salicylate. Intravenous sodium bicarbonate should be given to decrease tissue penetration and facilitate the excretion of salicylate through the kidneys unless the patient has oliguric AKI, pulmonary edema, or cerebral edema.

p0255 Despite being highly protein bound, saturation of protein binding in significant poisoning and its limited volume of distribution makes it amenable to removal by dialysis. ERCT is recommended in patients with severe salicylate poisoning reflected by altered mental state, acute respiratory distress syndrome, and failure to respond to medical management irrespective of salicylate levels. ERCT is also indicated, regardless of signs and symptoms, at levels greater than 100 mg/dL (>7.2 mmol/L). Lower thresholds are applied for patients with AKI. Intermittent HD has advantages compared with hemoperfusion because the former allows rapid correction of associated electrolyte abnormalities and acidemia. CKRT provides lower salicylate clearance than HD and should be implemented only if intermittent modalities are not available.²³

s0100 Theophylline

p0260 β_2 agonists have gradually replaced theophylline in the treatment of lung disease over the past 20 years. Still, new therapeutic uses are emerging, and it remains a potential toxin. Methylxanthine (MW 180 Da) overdose manifests with nausea, vomiting, diarrhea, gastrointestinal hemorrhage, hypokalemia, seizures, arrhythmias, and hypotension. Overdose may be acute or chronic. Theophylline is readily cleared by either HD or hemoperfusion because of its low volume of distribution and minimal protein binding. ERCT is recommended for severe acute theophylline poisoning (theophylline level >100 mg/L [>555 $\mu\text{mol/L}$], or the presence of seizures, life-threatening arrhythmias, shock, or failure of standard therapy). In chronic poisoning, extracorporeal removal is suggested if theophylline levels are greater than 60 mg/L (>333 $\mu\text{mol/L}$) or greater than 50 mg/L (>278 $\mu\text{mol/L}$) in a patient older than 60 years. Maximal theophylline clearance can be achieved with charcoal hemoperfusion. However, with the advent of new high-efficiency filters and higher achievable blood flow rates, the clearances achievable with HD now rival those with hemoperfusion and are associated with fewer side effects. Continuous venovenous HF can be used but requires a longer treatment duration than with hemoperfusion.²⁴

s0105 Valproate Acid

p0265 Valproate acid (VPA) is commonly used to treat epilepsy and bipolar disorder and to prevent migraine. VPA has a low molecular weight (144 Da), has a small volume of distribution (0.1–0.5 L/kg) and saturable plasma protein binding. At therapeutic concentrations, VPA is 94% protein bound, but it decreases to as low as 15% as concentrations rise to more than 1000 mg/dL, leading to greater clinical toxicity. Overdose may manifest with mild confusion, lethargy, nausea, vomiting, tachycardia, hypotension, metabolic acidosis, and electrolyte disturbances (hyponatremia and hypocalcemia). VPA concentrations can be helpful

in determining the severity of intoxication; mild if less than 400 mg/L, moderate if between 450 and 850 mg/L, and severe if more than 850 mg/L. In addition to electrolyte abnormalities (hypernatremia, hypocalcemia), evidence of impaired mitochondrial function can be present, including metabolic acidosis, hyperlactatemia, and hyperammonemia.

ERCT is recommended for severe overdose (VPA level >1300 mg/L [>9000 $\mu\text{mol/L}$], cerebral edema, or shock). ERCT can be considered for VPA concentrations greater than 900 mg/L (>6250 $\mu\text{mol/L}$), coma or respiratory depression requiring ventilation, acute hyperammonemia, or pH less than 7.1. HD is preferred to hemoperfusion because it clears the unbound drug and also reverses associated metabolic abnormalities.²⁵

Tricyclic Antidepressant Drugs

ERCT is not likely to have any clinical benefit for patients who have experienced an overdose with TCA drugs.²⁶

Thallium

Thallium salts were once used in the treatment of ringworm. Poisoning continues to be reported where it is used as a rodenticide and a contaminant of herbal and illicit drug products.²⁷ It is absorbed extensively through almost all routes of exposure and is widely distributed across multiple body compartments. Thallium is a small molecule (MW 204 Da) and does not bind to protein, but the efficacy of extracorporeal removal is limited by its large volume of distribution.

Toxicity is caused by thallium replacing potassium as a stimulator or inhibitor of a variety of intracellular electrochemical and enzymatic processes. ECTR removes around 3% of total body stores over 6 hours, with hemoperfusion seeming to be the most efficient modality. PD and plasmapheresis do not appear to deliver significance clearances. Based on a recent literature review, the EXTRIP workgroup strongly recommended extracorporeal removal for treatment of thallium poisoning, given the absence of alternative treatments.²⁷ Intermittent HD is preferable, but hemoperfusion and CKRT are valid alternatives.

Barbiturates

The use of barbiturates in clinical practice has largely been replaced by benzodiazepines because of the lower risk from overdose and an available antidote to reverse toxicity. Nonetheless, barbiturate overdose is still a significant cause of fatal poisoning, especially phenobarbital, which is still used clinically and in veterinary practice.

Barbiturates mainly act on the CNS, and symptoms of a moderate overdose typically include sluggishness, lack of coordination, slow speech, faulty judgment, and drowsiness. Shallow respiration and coma occur in severe poisoning. It is recommended that ERCT be restricted to cases of severe, long-acting barbiturate poisoning with clinical features including prolonged coma, respiratory depression requiring ventilation, shock, and a failure to treat effectively with multiple-dose activated charcoal. ERCT improves barbiturate elimination and will be most beneficial if it is initiated early, ideally within 24 hours of exposure. Intermittent HD is the preferred modality: it is equivalent to hemoperfusion and superior to CVVH, which is in turn superior to PD.²⁸

Acetaminophen/Paracetamol

Acute ingestions of acetaminophen (MW 46 Da) can still be lethal even when patients are treated early, within 8 hours of ingestion, and despite the availability of the antidote, *N*-acetylcysteine (NAC). In massive ingestions (>500 mg/kg), patients can present with an altered level of consciousness and metabolic acidosis with elevated lactate concentrations. Acetaminophen concentrations should be checked after 4 hours after ingestion or earlier in the case of massive ingestions.

p0305 There is a paucity of high-quality evidence for extracorporeal removal of acetaminophen, although it may have a role in the rare cases when *N*-acetylcysteine has not been effective. Patients with severe poisoning can also develop AKI even without liver dysfunction as a result of local production of the toxic by-product, *N*-acetyl-*p*-benzoquinone imine, by kidney prostaglandin synthetases. Thus, patients with acetaminophen-associated AKI may develop indications for ERCT other than acute toxin removal. Intermittent HD is the preferred modality.²⁹

s0130 Carbamazepine

p0310 Carbamazepine is currently used to treat bipolar disorder, neuropathic pain, hyperactivity, and seizure disorder. It has a low molecular weight (236 Da), a structure similar to that of TCAs, and is highly bound to both albumin and α 1-acid glycoprotein (70%–80%). It is metabolized in the liver, and only 1% to 3% of the drug is excreted unchanged in the urine. The therapeutic range is 4 to 12 mg/L, with toxicity usually occurring at more than 40 mg/L. Symptoms are mainly neurologic and include movement disorders, altered mental status, and seizures. Respiratory depression and cardiovascular symptoms can occur in severe overdose.

p0315 As the drug is moderately well dialyzed, extracorporeal removal is suggested in severe carbamazepine poisoning (as evidenced by coma,

respiratory depression requiring ventilation, or carbamazepine levels that do not respond to standard therapy with activated charcoal). Intermittent HD is the favored modality. If HD is not available, hemoperfusion or CKRT can be considered.³⁰

Phenytoin

Phenytoin (MW 252 Da) has a large volume of distribution and is extensively protein bound (90%) but dissociates easily and is amenable to extracorporeal removal. However, phenytoin poisoning rarely causes permanent organ damage or death, so extracorporeal removal is recommended only in very severe poisoning that has been resistant to standard therapy.³¹ Enteral absorption of phenytoin is slow and unpredictable, so vigilance should be maintained for the delayed effects of poisoning. Extracorporeal removal should be reserved for patients with severe poisoning or prolonged coma, aiming to reduce morbidity and resource use rather than decrease mortality.

Digoxin

Digoxin (MW 781 Da) has variable absorption from the gut depending on the size of the ingestion. Although it is 20% to 30% protein bound, it also has a large volume of distribution and an effective antidote (digoxin immune Fab), so extracorporeal removal is not useful for managing overdose.³²

SELF-ASSESSMENT QUESTIONS

- o0010 1. Which one of the following statements is *true* regarding ethylene glycol toxicity?
- o0015 A. Ethylene glycol is metabolized to glycolic acid.
- o0020 B. Ethylene glycol is metabolized to formic acid.
- o0025 C. Ethylene glycol is metabolized by alcohol hydrogenase.
- o0030 D. Urinalysis demonstrates uric acid crystals.
- o0035 E. Ethylene glycol toxicity is associated with a normal anion gap.
- o0040 2. Which one of the following is *not* associated with a raised osmolar gap?
- o0045 A. Ethylene glycol intoxication
- o0050 B. Methanol intoxication
- o0055 C. Isopropanol intoxication
- o0060 D. Diabetic ketoacidosis
- o0065 E. Acute kidney injury
- o0070 3. Intermittent HD is *not* effective for removal of which of the following in overdose?
- o0075 A. Lithium

- B. Ethylene glycol o0080
- C. Metoprolol o0085
- D. Carvedilol o0090
- E. Methanol o0095
4. Which one of the following statements is *true*? o0100
- A. Hemoperfusion remains the first-choice treatment for removal of theophyllines. o0105
- B. If endogenous clearance rates of toxins are less than 10 mL/min/kg body weight, ERCT should be considered. o0110
- C. Highly protein-bound toxins are not suitable for ERCT. o0115
- D. When continuous venovenous hemofiltration is used solely for poison removal, conventional doses for patients with AKI should be used. o0120
- E. Toxin rebound after intermittent HD may be mitigated by frequent scheduling of treatment or adjunctive therapy with one of the continuous renal replacement therapies. o0125

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